

Optimizing Stroke Prediction Models for Real-World Application: Tackling Data Imbalance Without Synthetic Samples

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Research area: Stroke prediction is a critical area in healthcare, often hindered by imbalanced datasets. Previous approaches [1] have relied on synthetic data generation techniques like SMOTE to address this imbalance. However, synthetic health data may lead to unreliable real-world performance. Moreover, accuracy and other weighted measures hide the underrepresentation of the minority class in contemporary work.

Goals: This study explores alternative methods to effectively classify stroke patients using imbalanced data without synthetic data generation. Furthermore, this study argues that recall is the proper metric for evaluation.

Methods: We employed a dataset of 5110 patients (4861 normal, 249 strokes) with 11 features. This study leveraged multiple machine learning models for the purpose of stroke classification. Emphasis was put on reducing the number of false negatives through proper consideration of the medical nature of the problem. Contemporary work based on this stroke prediction dataset has focused on the use of synthetic data by means of SMOTE. Considering the risk of synthetic samples not representing the minority class in real-world populations, this study excludes such methods [2]. Our approach involves the use of multiple models including Gaussian Naïve Bayes with custom priors, Artificial Neural Network (ANN), Logistic Regression with standardization and custom weights, SVM with custom weights, and Ensemble Methods using Soft Voting with multiple combinations of models. Furthermore, we use a Random Forest classifier with Balanced Class Weights to prove the inefficacy of accuracy, and weighted measures as a metric to judge a model in case of severely imbalanced data.

Results: The Random Forest classifier showed high values in accuracy, and weighted measures while not being able to correctly predict a single stroke case. This proves the inefficacy of accuracy, and weighted measures as a metric to judge a model in case of severely imbalanced data. Our models achieved high recall values for stroke class without synthetic data generation. The three best models are: Logistic Regression with standardization and custom weights, Ensemble using Soft Voting with custom weights (GNB and LR with standardization and balanced weights), and Support Vector Machine with custom weights. The models respectively attained recall values of 92%, 92% and 94%.

Conclusion: This study demonstrates that high-performing stroke classification models can be developed using imbalanced datasets without resorting to synthetic data generation. In stroke prediction, false negatives are far costlier than false positives. The proposed method involves the minimization of false negatives and so suggests the use of recall to evaluate the performance of machine learning models. The best models obtain high recall values of 92%, and 94%. These findings suggest that effective stroke prediction is possible in real-world scenarios where data imbalance is common, potentially improving early diagnosis and treatment of stroke patients.

Keywords: Imbalanced Dataset Classification, Stroke Prediction, Ensemble ML Models, Recall

References:

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